OP-AMP Frequency response and Mathematical tools

Aim

a) To study the frequency response of OP-AMP inverting and non-inverting amplifiers using IC 741.

b) To construct mathematical tools (adder, subtractor, differentiator and integrator) using IC 741.

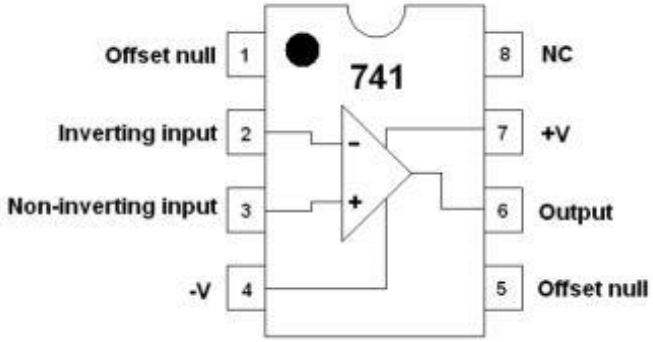
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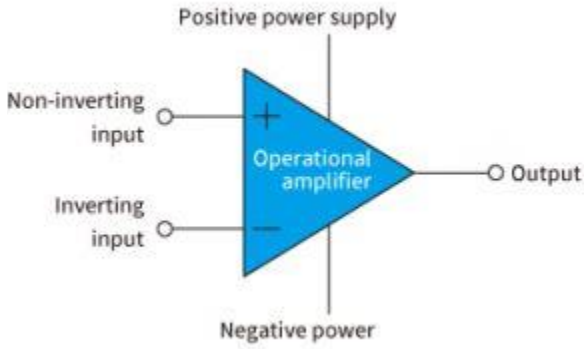
Apparatus

IC 741, Resistors, CRO, breadboard, DC regulated fixed power supply, Regulated DC power supply, function generator, digital multimeter, connecting wires.

Theory

OPAMP is a direct-coupled high-gain differential-input amplifier that can amplify weak electric signals. An operational amplifier has two input pins and one output pin. Its basic role is to amplify and output the voltage difference between the two input pins. The name "operational amplifier" comes from the fact that they were originally used to perform mathematical operations such as adder, subtractor, integration, differentiation, etc. The advantage of OP-AMP is that negative feedback can be applied. The performance of the OP-AMP with negative feedback is controlled by the feedback elements independent of the characteristics of the transistors, capacitors and resistors that constitute the OPAMP. As the feedback elements are usually passive, the circuit operation is very stable and predictable.

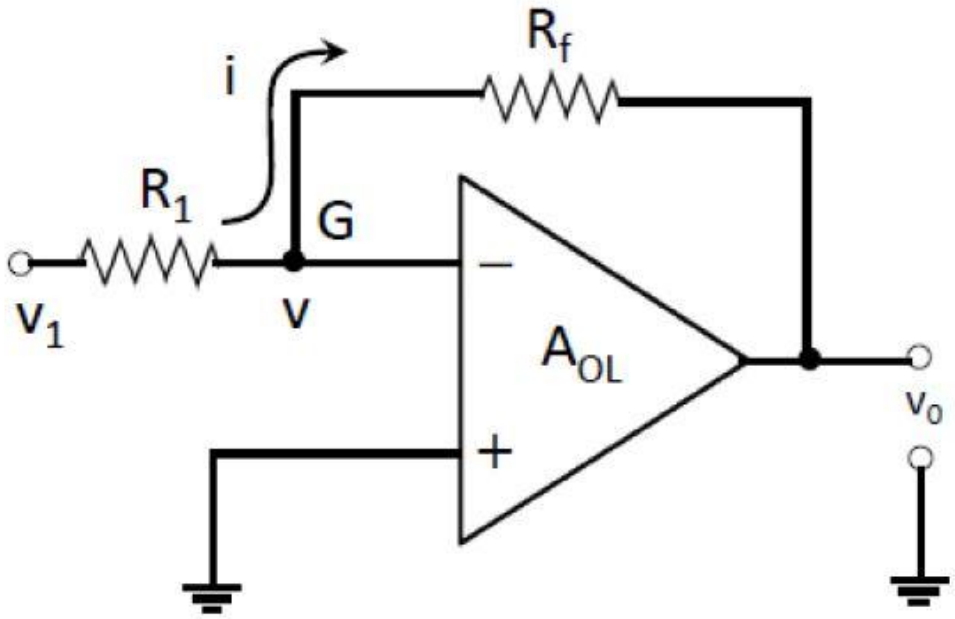




Inverting Amplifier

An inverting amplifier is a type of operational amplifier circuit which produces an output which is out of phase with respect to its input by 180˚.

This means that if the input pulse is positive, then the output pulse will be negative and vice versa.



Open loop gain (A_{OL}) $\approx \infty$

v_o finite due to –ve feedback

$v = (v_o/A_{OL}) \sim 0$ (virtual ground)

Current (i) in $R_1 = (v_1\text{-}v)/R_1$

For ideal Op amp, that 'i' passes through R_f also, $i = (V\text{-}V_o)/R_f$

As G is a virtual ground, $v \approx 0$.

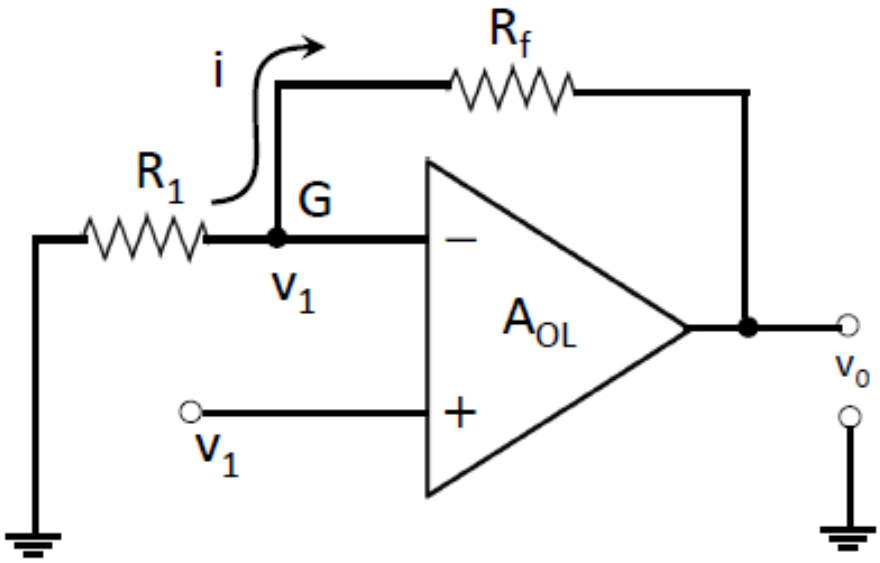
So, $v_1/R_1 = -(v_o/R_f)$ $\quad v_o/v_1 = -(R_f/R_1)$

The voltage gain (Closed loop gain) of the **inverting operational amplifier** is,

$$A_v = -\frac{R_f}{R_i}$$

This indicates that the voltage gain of the inverting amplifier is decided by the ratio of the feedback resistor to the input resistor with the minus sign indicating the phase-reversal (180^0 phase shift). Further, it is to be noted that the input impedance of the inverting amplifier is nothing but R_i. Inverting amplifiers exhibit excellent linear characteristics which make them ideal as DC amplifiers. Moreover, they are often used to convert input current to the output voltage in the form of Transresistance or Transimpedance Amplifiers. Further, these can also be used in audio mixers when used in the form of Summing Amplifiers.

Non-inverting amplifier



Non inverting amplifier is an op amp based amplifier with positive voltage gain.

The input voltage vi is applied to the non-inverting terminal.

Similar to the discussion of inverting amplifier, voltage at G is found to be v1

$$\frac{v_o - v_1}{R_f} = \frac{v_1}{R_1}$$

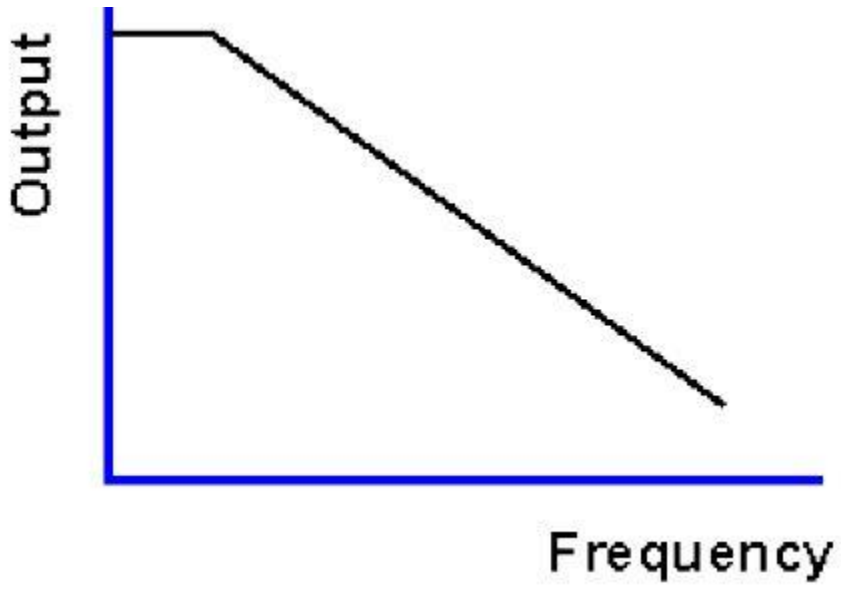
$$\frac{v_o}{R_f} = v_1\left(\frac{1}{R_f} + \frac{1}{R_1}\right)$$

$$\frac{v_o}{R_f} = \frac{v_1}{R_f}\left(1 + \frac{R_f}{R_1}\right)$$

$$\frac{v_o}{v_1} = 1 + \frac{R_f}{R_1}$$

Frequency response

The frequency response of a typical op amp chip will often start to fall at a very low frequency when operated in its open loop mode.



Open loop op amp frequency response

The point at which the frequency starts to roll off is known as the break point [typically the -3dB point is known as the break point].

Most op amps have their bandwidth limited as a result of what is called compensation which is added.

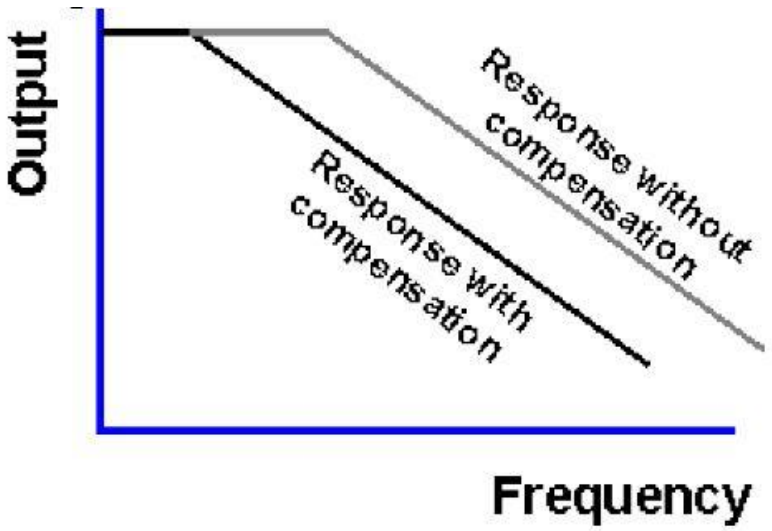
OP-AMP frequency compensation:

Most op amp chips have frequency compensation added to them. It is introduced to ensure they remain stable and do not produce unwanted high frequency spurious oscillations.

It is required because stray capacitances in the chip can cause unwanted phase shifts at high frequencies, e.g. 1 MHz and more. While the stray capacitance levels may not be significant at low frequencies, they can cause significant problems at higher frequencies, and they are almost impossible to eliminate in the chip.

Most op amp IC manufacturers solve this problem by intentionally reducing the open-loop gain at high frequencies. This is called compensation and it is normally implemented by bypassing one of the internal amplifier stages with a high-pass filter. The aim is to reduce the gain to less than unity at frequencies where there could be a possibility of oscillation.

Very early op amps did not have this frequency compensation built into the chip and external compensation components were required on the pins provided - the 709 was a prime example of this. Later chips such as the 741 had internal compensation making the chips much easier to use. However they also had a low open loop break point. In the case of the 741 it was just 10 Hz. This compensation is now standard in all general purpose op amp chips. Modern chips continue to have it built in a standard.



Op amp frequency response with and without frequency compensation

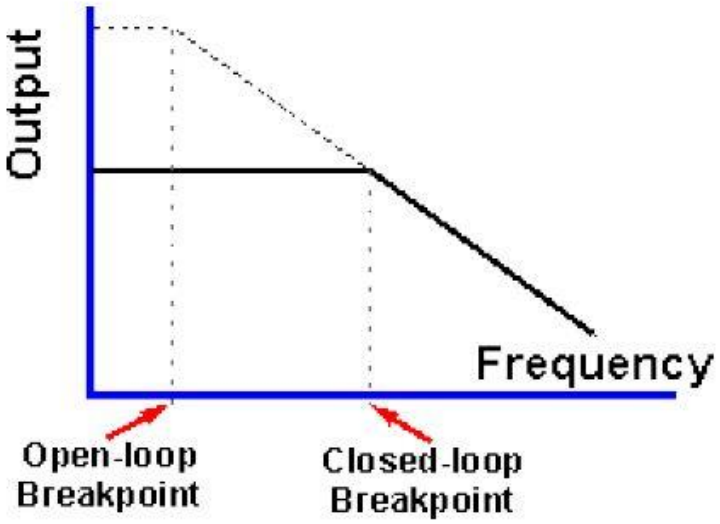
Frequency compensation is the major reason why op-amps are not very fast devices - the higher frequency components of the signals are intentionally attenuated. The frequency at which the Op amp open loop gain falls to unity, is called fT - as for bipolar transistors. This frequency gives a good indication of the speed of the op-amp.

However, comparators do not use negative feedback and as a result they are designed without compensation and their speed of operation is typically much faster than that of op amps.

Feedback vs bandwidth:

In view of the very high gain of the operational amplifier it is possible to, in effect, exchange some of the open loop gain for bandwidth.

For a circuit like this, applying feedback will reduce the gain but increase the bandwidth.



Closed loop op amp frequency response

Op amp gain bandwidth product:

When designing an op amp circuit, a figure known as the op amp gain bandwidth product is important.

The op amp gain bandwidth product is generally specified for a particular op amp type an open loop configuration and the output loaded:

$$\text{Gain bandwidth product} = A_V \times f$$

where:

Av = Voltage gain,

f = cut off frequency (Hz).

The op amp gain bandwidth product is constant for voltage-feedback amplifiers. However it is not applicable for current feedback amplifiers because relationship between gain and bandwidth is not linear.

Therefore decreasing the gain by a factor of ten will increase the bandwidth by the same factor.

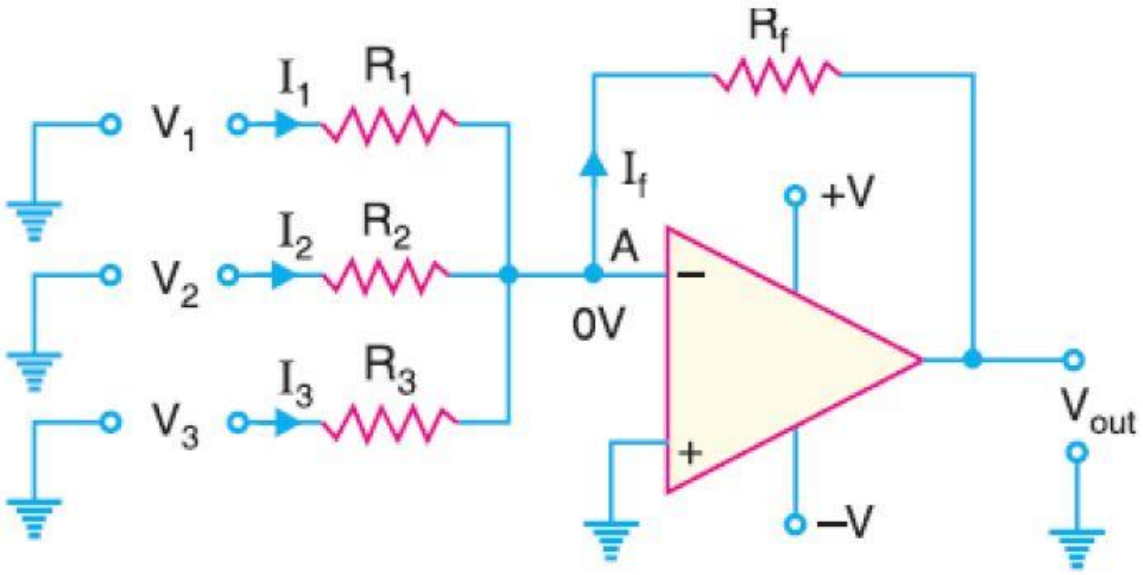
Op-amp Characteristics:

An ideal op amp is usually considered to have the following characteristics:

- Infinite open-loop gain $G = v_{out} / v_{in}$
- Infinite input impedance R_{in}, and so zero input current
- Zero input offset voltage
- Infinite output voltage range
- Infinite bandwidth with zero phase shift
- Infinite slew rate
- Zero output impedance R_{out}
- Zero noise
- Infinite common-mode rejection ratio (CMRR)
- Infinite power supply rejection ratio.

Mathematical tools - (adder, subtractor, differentiator and integrator)

(i) Analog Circuits Adding or Adder or Summing Amplifier:



$$I_f = I_1 + I_2 + I_3$$

$$\text{Output voltage, } V_{out} = -I_f R_f = -R_f(I_1 + I_2 + I_3)$$

$$= -R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)$$

$$V_{out} = -R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)$$

If $R_1 = R_2 = R_3 = R$, then, we have,

$$V_{out} = -\frac{R_f}{R}(V_1 + V_2 + V_3)$$

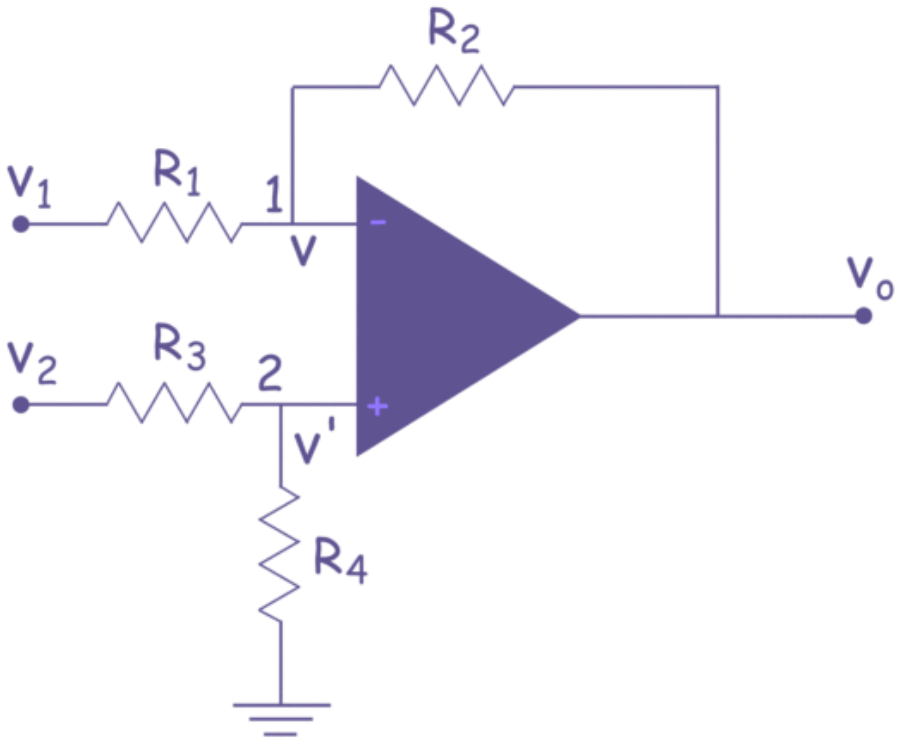
$$R_f = R_1 = R_2 = R_3$$

$$V_{out} = -(V_1 + V_2 + V_3)$$

Thus, the output voltage is the *inverted* algebraic sum of the input voltages. Hence the circuit is termed a *summing amplifier* or an *adder*.

(ii) Subtractor or difference Amplifier:

A difference amplifier or Op amp subtractor is a specially designed op amp based amplifier circuit, which amplifies the difference between two input signals and rejects any signals common to both inputs.



Let us consider the above op amp circuit. Now, by applying Kirchhoff Current Law at node 1, we get,

$$\frac{v_1 - v}{R_1} = \frac{v - v_0}{R_2}$$

We have written this equation by assuming that there is no current entering in the inverting terminal of the op amp.

Now, by simplifying the above equation, we get,

$$\frac{v_1}{R_1}+\frac{v_0}{R_2}=\frac{v}{R_1}+\frac{v}{R_2}$$

$$\Rightarrow \frac{v_0}{R_2}=\frac{v}{R_1}+\frac{v}{R_2}-\frac{v_1}{R_1}$$

$$\Rightarrow v_0=\left(\frac{R_2}{R_1}+1\right)v-\frac{R_2}{R_1}v_1 \Rightarrow v_0=\left(1+\frac{R_2}{R_1}\right)v-\frac{R_2}{R_1}v_1 \quad \ldots\ldots(i)$$

Now, by applying Kirchhoff Current Law, at node 2, we get,

$$\frac{v_2-v'}{R_3}=\frac{v'-0}{R_4} \Rightarrow v_2=v'\left(1+\frac{R_3}{R_4}\right)$$

$$\Rightarrow \frac{v_2}{R_3}=\frac{v'}{R_4}+\frac{v'}{R_3} \Rightarrow v'=v_2\left(\frac{R_4}{R_3+R_4}\right) \quad \ldots\ldots(ii)$$

We know that, in ideal op amp, voltage at inverting input is same as the voltage at non inverting input. Hence,

$$v=v'$$

So, now from equation (i) and (ii), we get,

$$v_0=\left(1+\frac{R_2}{R_1}\right)\frac{R_4}{R_3+R_4}v_2-\frac{R_2}{R_1}v_1$$

$$\Rightarrow v_0=\frac{\left(1+\frac{R_2}{R_1}\right)}{\left(1+\frac{R_3}{R_4}\right)}v_2-\frac{R_2}{R_1}v_1$$

$$v_0=\frac{R_2}{R_1}\cdot\frac{1+\frac{R_1}{R_2}}{1+\frac{R_3}{R_4}}v_2-\frac{R_2}{R_1}v_1 \cdots\cdots\cdots (iii)$$

The **difference amplifier** must reject any signal common to both inputs. That means, if polarity and magnitude of both input signals are same, the output must be zero.

$$\because \ when,\ v_1=v_2,\ then,\ v_0=0$$

This condition must be satisfied only when,

$$\frac{R_1}{R_2}=\frac{R_3}{R_4} \Leftrightarrow R_1 \vdots R_2 = R_3 \vdots R_4$$

In that case, equation (iii) becomes,

$$v_0 = \frac{R_2}{R_1} v_2 - \frac{R_2}{R_1} v_1$$

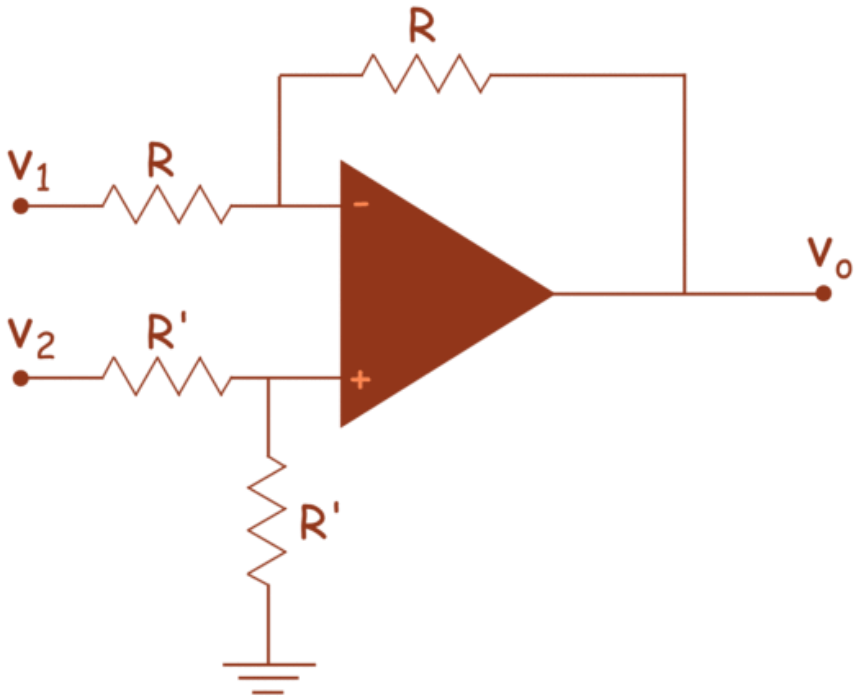
$$\Rightarrow \; v_0 = \frac{R_2}{R_1}(v_2 - v_1) \quad \ldots\ldots (iv)$$

Again, if we make, $R_1 = R_2$, then equation (iv) becomes,

$$v_0 = v_2 - v_1$$

So, if $R_1 = R_2$ and also $R_3 = R_4$ then the difference amplifier becomes a perfect subtractor, which subtracts directly the input signals.

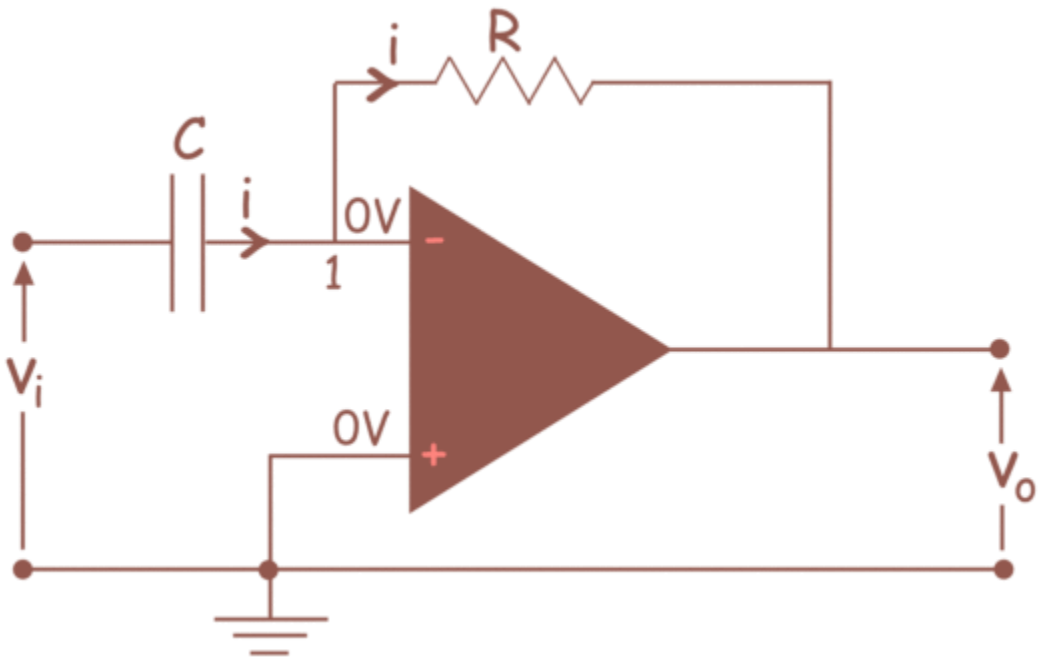
Finally, the circuit of **op amp substractor** becomes,



(iii) Differentiator:

Differentiator is an op-amp based circuit, whose output signal is proportional to differentiation of input signal

An op amp differentiator is basically an inverting amplifier with a capacitor of suitable value at its input terminal. The figure below shows the basic circuit diagram of an opamp differentiator.



We will first assume that the op amp used here is an ideal op amp. We know that the voltage at both inverting and non inverting terminals of an ideal op amp is same. As the electric potential at non inverting terminal is zero since it is grounded. The electric potential of inverting terminal is also zero, as the opamp is ideal. Because, we know that the electric potential at non – inverting and inverting terminals. It is also known to us that the current entering through inverting and non inverting terminal of an ideal op amp is zero. Considering, these conditions of an ideal op amp, if we apply Kirchhoff Current Law at node 1 of the op amp differentiator circuit, we get,

$$C\frac{dv_i}{dt} = -\frac{v_0}{R}$$

$$\Rightarrow v_0 = -RC\frac{dv_i}{dt}$$

ie,

$$q = Cv_1 \quad \rightarrow \quad v_1 = \frac{q}{C} \qquad (1)$$

$$\frac{dv_1}{dt} = \frac{1}{C}\frac{dq}{dt} = \frac{i}{C} \quad (2)$$

$$i = -\frac{v_o}{R} \quad (3)$$

(2) + (3) → (4)

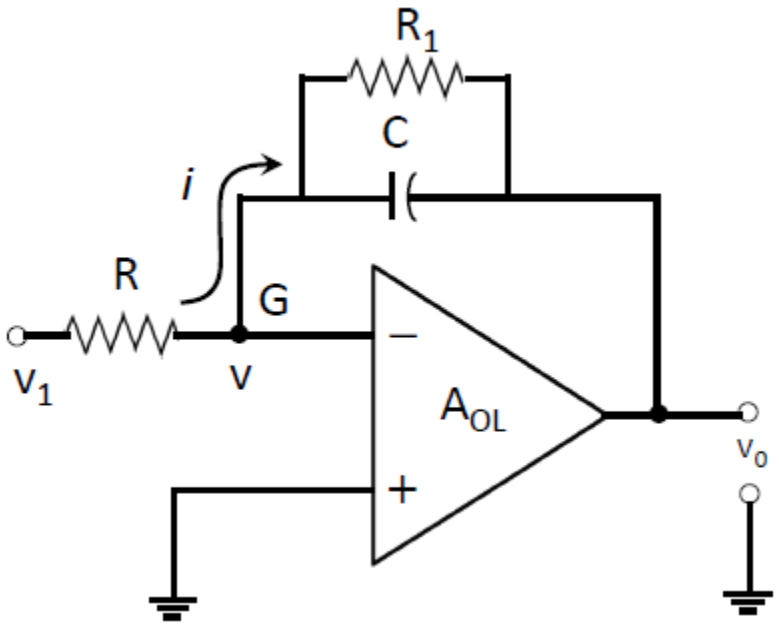
$$v_o = -RC\frac{dv_1}{dt} \quad (4)$$

Scale factor = - RC

The above equation shows that the output voltage is the derivative of the input voltage.

(iv) Integrator

An integrator is an op-amp circuit, whose output is proportional to the integral of input signal.



Current in R, $$i = \frac{v_1}{R}$$

Current flows through the feedback capacitor C to produce output voltage v_o,

$$v_o = -\frac{q}{C} = -\frac{1}{C}\int_0^t i\,dt$$

$$= -\frac{1}{RC}\int_0^t v_1\,dt$$

This circuit has a drawback. As the capacitor is an open circuit for DC and the DC gain of the OP AMP circuit is infinite, any DC voltage at the input would drive the OP AMP output to saturation. To avoid this possibility, a resistance R1 is connected in parallel with the capacitance C, which limits the dc gain of the circuit. The value of R1 is so chosen that,

$$R_1 \gg \left|\frac{1}{\omega C}\right|$$

where 'ω' is the angular frequency of the input signal.

Because of the negative sign, the circuit is actually an *inverting integrator*.

Scale factor = -1/(CR).

RESULT

(a) The inverting and non-inverting amplifiers were set up using IC 741 and their frequency responses were studied.

(b) The circuits for adder, subtractor, differentiator and integrator were set up using IC 741 and the waveforms were observed on the CRO.

References:

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